

Prototype Tasks

Phase 0 — Project Setup

- ☐ Create backend/frontend structure
 - ☐ Create Python environment
 - ☐ Install core dependencies
 - ☐ Create configuration system
 - ☐ Create basic FastAPI server
 - ☐ Create basic frontend
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Phase 1 — Transportation Model

- ☐ Implement Node model
- ☐ Implement Edge model
- ☐ Implement Vehicle model
- ☐ Implement Job/Customer model
- ☐ Implement Traffic model
- ☐ Generate synthetic urban graph
- ☐ Generate vehicles and jobs
- ☐ Add depot and capacity constraints

Deliverable

A reproducible transportation scenario that can be visualized.

Phase 2 — Routing Problem

- ☐ Implement route representation
- ☐ Implement route decoding
- ☐ Implement capacity validation
- ☐ Implement route-duration validation
- ☐ Implement travel-time calculation
- ☐ Implement distance calculation
- ☐ Implement congestion cost
- ☐ Implement complete fitness function

Deliverable

Given a candidate route, the system can calculate its total cost and feasibility.

Phase 3 — Baseline Optimizer

- ☐ Implement Greedy / Nearest Neighbour
- ☐ Generate valid routes
- ☐ Calculate baseline metrics

Deliverable

A working baseline solution.

Phase 4 — QPSO

- ☐ Define solution encoding
- ☐ Implement population initialization
- ☐ Implement fitness evaluation
- ☐ Implement QPSO update mechanism
- ☐ Implement best-solution tracking
- ☐ Implement convergence history
- ☐ Implement constraint handling
- ☐ Add random seed support

Deliverable

QPSO produces feasible vehicle routes and convergence data.

Phase 5 — Classical Comparison

- ☐ Implement PSO or GA
- ☐ Ensure common optimizer interface
- ☐ Run same problem across algorithms
- ☐ Record metrics

Deliverable

Comparable results for:

Greedy
PSO/GA
QPSO

Phase 6 — Dynamic Traffic Demo

- ☐ Add traffic incident simulation
- ☐ Modify selected edge weights
- ☐ Display congested road
- ☐ Recalculate route
- ☐ Run optimization again
- ☐ Display old vs new route
- ☐ Calculate improvement/difference

Deliverable

Working:

Traffic Change → Re-optimization → New Route

demonstration.

Phase 7 — Benchmarking

- ☐ Create benchmark runner
- ☐ Run multiple algorithms
- ☐ Measure runtime
- ☐ Measure solution cost
- ☐ Measure travel time
- ☐ Measure distance
- ☐ Record convergence
- ☐ Calculate optimality gap where possible
- ☐ Generate comparison charts

Deliverable

A reproducible benchmark report.

Phase 8 — Demo UI

- ☐ Build transportation map
- ☐ Display vehicles
- ☐ Display customer locations
- ☐ Display routes
- ☐ Display traffic conditions
- ☐ Add optimization controls
- ☐ Add traffic incident button

- ☐ Add metric cards
- ☐ Add algorithm comparison
- ☐ Add convergence chart

Deliverable

A judge-facing interactive prototype.

Final Demo Checklist

Before presenting, verify:

- ☐ Scenario can be generated reliably
 - ☐ Routes are feasible
 - ☐ QPSO runs successfully
 - ☐ Baseline runs successfully
 - ☐ Traffic incident changes edge costs
 - ☐ Re-optimization changes route when appropriate
 - ☐ Metrics are calculated from actual runs
 - ☐ Benchmark results are reproducible
 - ☐ No fake performance numbers
 - ☐ Demo can be completed in 3–5 minutes
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Demo Script

1. Show the transportation network.
2. Explain:
"We have multiple vehicles and delivery points operating under different traffic conditions."
3. Run optimization.
4. Show the generated routes and metrics.
5. Trigger a traffic incident.
6. Explain:
"The cost of this road has increased significantly."
7. Re-optimize.
8. Show the changed route.

9. Open benchmark panel.

10. Compare:

Greedy vs PSO/GA vs QPSO.

11. Explain:

"We are not assuming QPSO is better.

We are measuring solution quality,
runtime, convergence and scalability."

12. Conclude:

"This prototype validates the core optimization
loop. The next phase is the complete PRD
and rigorous algorithm specification."